

Course Title	Digital Logic and Computer Organization
Credit Structure	3-0-2-4
Course Code	IC121
Program	B.Tech. (ICT)
Category	Core

Course Objective

This course provides an introduction to the design and implementation of digital circuits and microprocessors. Topics include transistor network design, Boolean algebra, combinational circuits, sequential circuits, finite state machine design, processor pipelines, and memory hierarchy. Design methodology using both discrete components and hardware description languages is covered in the course.

Course Content

Combinational Circuits: Small and large Designs, Logic Expressions, Sum of Product Expression & Product of Sum Expression, Canonical Expression, Min-Terms, Max-Terms, Logic Minimization, Karnaugh Map, K-Map Minimization, Logic Minimization Algorithm, Minimization Software Other Gates, Buffer, Tri-State Buffer, Full Adder, Multiplexer, Decoder, Encoder, Circuit Timing Diagram, Signal Propagation Delay, Fan-In and Fan-Out, Programmable Logic Devices, Design Flow, Hardware Description Languages, Floating point standard Sequential Circuits: Core Modules, Small and large Designs. Latches, flipflops, Registers, HDL models, FSM, Single cycle, multi cycle, pipelining, Multipliers Memory. Multiplayers of memory, Memory types, Design Example: Multiprocessor Memory Architecture, HDL Models. Instruction Set Architecture. Types of Instruction Set Architecture, Design Example, Advanced Processor Architectures Computer Architecture: Interconnection, Memory Controller, I/O Peripheral Devices, Controlling and Interfacing I/O Devices, Data Transfer Mechanisms, Interrupts, Design Example: Interrupt Handling CPU. Computer Architecture: Security

Text Books:

1. Digital Design - M Mano and M Ciletti [Pearson]
2. Digital Design and Computer Architecture, Harris, Harris, 2nd Edition, Morgan Kaufmann.

Evaluation Components

In semester evaluation – 40%

End semester evaluation – 40%

Lab evaluation – 20%

Program Outcomes

At the end of the course student will be able to

- Apply the concepts in Boolean algebra and derive switching expressions for logic circuits.
- Design various combinational circuits for arithmetic and logical operations.
- Apply the concepts of digital storage elements to design synchronous and asynchronous sequential circuits.
- Transistor level implementation of logic circuits.
- Implementation of digital circuits on hardware kits using CMOS ICs.
- Understand the internal architecture of the CPU, memory hierarchy, assembly language instructions and addressing modes and timing analysis of instructions

P1	P2	P3	P4	P5	P6	P7	P8	P9	P10	P11	P12
X		X		X	X						X